Weak endpoint covariance and first-order stability of shrinking surgery

27 September 2026. Continuation of `e75105d`, not an export package. The inputs and curvature conventions are those of I40/I41 and I51.

The main construction-side conclusion is Theorem 9: on a suitably chosen complex-hyperbolic shrinking-surgery recovery sequence, every vanishing-amplitude perturbation in either of two specified TT or Bianchi slices has first-order coefficient at least four. No rate relative to h, derivative bound, Sobolev constant, or competitive endpoint energy is assumed. The proof retains the classical positive Einstein operator on the exterior and charges cutoff errors only to the surgery budget. Its nonlinear lower bound also identifies the equality condition inside these slices. Sections 3–4 derive this from the actual construction; Theorem 7 records the weaker estimate which does not use the exterior gap.

The endpoint calculation in Sections 1–2 separately identifies the next derivative controls and measures the phase curvature they still miss. Neither the unrestricted weak zero-end gap nor the all-metric optimal coefficient is proved here.

1. The weighted second derivative of the determinant field

Work with the same actual constant-weight solution as I51, with scalar curvature minus four and class square at least T_+. Throughout this section all integral norms are normalized by V. Write

$$\rho = |\Phi|^2 \leqslant 2, \quad \Pi = \nabla_A \Phi, \quad H = \nabla_A \Pi, \quad \zeta = \langle \Phi, \Pi \rangle, \quad \gamma = \operatorname{Im} \zeta,$$
$$\rho_* = 4/3, \quad m_* = 128/81, \quad \tau = t_\Phi, \quad m = |\tau|^2, \quad S = m_* + 2\rho^2 |\Pi|^2, \quad w = \rho^4/S, \quad \eta = r + D_\beta/V + \beta^{-2} \leqslant 1.$$

The real ASD forms are $f = -iF_A^-$ and $Q = -iq_-(\tau)$, with

$$\rho^2 \Delta_H f + \rho^3 f = -2Q + R, \qquad |R| \leqslant C(\rho^3 |\operatorname{Ric}^\circ| + \rho^{3/2} |\zeta| |\Pi|), \qquad |Q|^2 = m^2/8. \tag{1.1}$$

We use the following already proved, same-solution estimates from I51 Section 2.2; no identification with an earlier variable-weight solution is made:

$$\fint \frac{(S - 2m_*)^2}{S} \leqslant C\sqrt{\eta}, \qquad K = -\frac{4m_*}{\rho_*^6} \frac{\rho^3 Q}{S}, \qquad \|K\|_\infty \leqslant C, \qquad \fint |f - K|^2 \leqslant C\sqrt{\eta}. \tag{1.2}$$

Also $m \leqslant S$, $S \geqslant m_* > 0$,

$$\fint (|\rho - \rho_*|^2 + |\zeta|^2 + |\operatorname{Ric}^\circ|^2) \leqslant C\eta, \qquad \fint (|\Pi|^4 + |H|^2 + |\nabla \tau|^2) \leqslant C. \tag{1.3}$$

Lemma 1. Under these hypotheses,

$$\boxed{\fint w\{|\Delta_H f|^2 + |\Delta \rho|^2\} \leqslant C\sqrt{\eta}.} \tag{1.4}$$

Proof. Put $\lambda = 4m_*/\rho_*^6$. Equation (1.1) gives

$$\rho^2\Delta_H f = -\rho^3(f-K) - (2-\lambda\rho^6/S)Q + R.$$

Divide its squared norm by S. The first term is controlled by (1.2). For the second use

$$2-\lambda\rho^6/S = 2(S-2m_*)/S + \lambda(\rho_*^6-\rho^6)/S,$$

and $m \leqslant S$: its contribution is at most

$$C\frac{(S-2m_*)^2}{S} + C|\rho-\rho_*|^2.$$

Finally $|R|^2/S \leqslant C(|\operatorname{Ric}^\circ|^2 + |\zeta|^2)$, because $\rho^2|\Pi|^2/S \leqslant 1/2$. This proves the first bound. The scalar equation is

$$\Delta\rho = -2|\Pi|^2 + 2\rho - \rho^2, \qquad \rho^2\Delta\rho = -(S-2m_*) + (2\rho^3 - \rho^4 - m_*).$$

The last polynomial vanishes at ρ_*. Squaring, dividing by S, and using (1.2)–(1.3) proves the second bound. QED.

This is a consequence of the existing saturation identities, not an independent endpoint exclusion. In particular w is not bounded below on the entire manifold.

2. A precise remaining current–Hessian correlation

The current is coclosed: $\delta\gamma = 0$. This follows by taking the imaginary part of the divergence of $\langle\Phi, \nabla_A\Phi\rangle$: the rough Laplacian of Φ is a real scalar multiple of Φ. Since $\delta f = -\gamma$, $df = *\gamma$,

$$\Delta_H f = -2(d\gamma)^-. \tag{2.1}$$

For a real coclosed one-form and a smooth nonnegative weight, integration by parts and $d(\gamma\wedge d\gamma) = d\gamma\wedge d\gamma$ give the exact formula

$$\begin{aligned}\int w|\nabla\gamma|^2 = 2\int w|(d\gamma)^-|^2 + \int w|\gamma|^2 - \int w\operatorname{Ric}^\circ(\gamma,\gamma)\\ - \int\langle dw, \nabla_{\gamma^\sharp}\gamma\rangle - \int dw\wedge\gamma\wedge d\gamma.\end{aligned} \tag{2.2}$$

Only Ricci, not W^-, enters this one-form identity. Differentiation of the particular weight gives

$$dw = \frac{4\rho^3(m_* + \rho^2|\Pi|^2)}{S^2}d\rho - \frac{4\rho^6}{S^2}\operatorname{Re}\langle\Pi, H\rangle. \tag{2.3}$$

Define the dimensionless nonnegative correlation

$$\mathscr{B} = \fint \frac{\rho^8|\Pi|^2|\zeta|^2}{S^3}|H|^2. \tag{2.4}$$

Proposition 2. The same actual solution satisfies

$$\boxed{\oint \frac{\rho^4}{S}|\nabla\zeta|^2 \leqslant C\sqrt{\eta} + C\mathscr{B}.} \tag{2.5}$$

Proof. The first term in (2.2) is small by (1.4) and (2.1). The Ricci term is $O(\sqrt{\eta}V)$, by the small L^2 Ricci error and the bounded L^4 norm of γ. Young's inequality absorbs half of the gradient term from the two weight errors. The remaining cost is bounded by $C\int |\gamma|^2|dw|^2/w$. By (2.3), the part containing $d\rho$ is at most $C|d\rho|^2$, using $|\gamma|^2 \leqslant \rho|\Pi|^2$ and

$$\frac{|dw|^2}{w} \leqslant C\frac{\rho^2}{S}|d\rho|^2 + C\frac{\rho^8|\Pi|^2}{S^3}|H|^2. \tag{2.6}$$

The other part is $CV\mathscr{B}$. Thus (2.5) holds for $\operatorname{Im}\zeta$. The weighted scalar Bochner formula is

$$\begin{aligned}\int w|\nabla^2\rho|^2 = {} & \int w(\Delta\rho)^2 - \int w\,\mathrm{Ric}(d\rho, d\rho) \\ & - \int (\Delta\rho)\langle dw, d\rho\rangle - \int \langle dw, \nabla_{\nabla\rho} d\rho\rangle.\end{aligned} \tag{2.7}$$

Use (1.4), then (2.6), $|d\rho| \leqslant 2|\zeta|$, and the same absorption. This proves the assertion for $\operatorname{Re}\zeta = d\rho/2$. At zeros, (2.6) is used through its nonsingular displayed upper bound; all original weights are smooth. Equivalently one may first use $w + \epsilon$ and pass to the limit. QED.

The coefficient of $|H|^2$ in (2.4) is uniformly bounded and tends to zero in measure, but this does not make $\mathscr{B}$ small from a mere L^2 Hessian bound. Uniform integrability of $|H|^2$ relative to normalized volume would suffice; the strictly weaker needed condition is the correlation $\mathscr{B} \to 0$ itself. Neither is added to the original input. On each region $\rho \geqslant c > 0, S \leqslant L$, (2.5) would then give strong current-derivative control. It would still not, by itself, give zero-end exclusion or control the torsion index on tiny cores.

The derivative which is missing from the classical argument should not be misidentified. In Armstrong's thesis, Sections 3.3.1–3.3.3, the full first torsion projection is combined with **pointwise constant torsion length**, and then a second Ricci identity and opposite-chirality Bianchi. The full first projection has already been proved with error in I41, Section 14. Average length convergence is not the pointwise derivative condition. The local actual cores in I50/I51 also prevent replacing this missing step by a claim that small quadratic derivative energy kills their index.

2.1 The normalized Hopf form still contains a length derivative

Put $j_\tau = \operatorname{Im}\langle\tau, \nabla_{3A}\tau\rangle$ and $a = \rho^3/S$. In the Clifford conventions of I41,

$$\delta Q = -\tfrac{1}{2}j_\tau + E_\tau, \qquad |E_\tau|^2 = \tfrac{1}{4}m|D^-_{3A}\tau|^2. \tag{2.8}$$

This follows directly by differentiating the quadratic Hopf map: at a point with $\tau = \sqrt{m}(1,0)$, write $D\tau = (A, B)$. The error one-form has components $\frac{\sqrt{m}}{2}(-\operatorname{Im}A, -\operatorname{Re}A, -\operatorname{Im}B, \operatorname{Re}B)$. Therefore the same $K = -\lambda aQ$ as in (1.2) satisfies

$$\delta K = \frac{\lambda}{2}\{a j_\tau + 2\iota_{\nabla a} Q\} - \lambda a E_\tau, \qquad \fint |aE_\tau|^2 \leqslant C\eta. \tag{2.9}$$

For the last estimate insert I51's actual torsion Dirac equation,

$$|D_{3A}^- \tau| \leqslant C|\zeta||\Pi| + C\rho^{3/2}|W^+ - W_0|,$$

and use $m \leqslant 2\rho^2|\Pi|^2$, $S \geqslant m_*$. The coefficients of $|\zeta|^2$ and $|W^+ - W_0|^2$ after multiplication by $a^2 m$ are uniformly bounded.

The first term of (2.9) is not automatically small. Even in the exact principal projection with ρ constant, the two almost-complex structures of opposite orientation give

$$j_\tau = -\tfrac{1}{2} J_+ dm, \qquad Q = -m\Omega_-/4, \qquad a j_\tau + 2\iota_{\nabla a} Q = -\tfrac{a}{2}\Big(J_+ - \tfrac{m}{S} J_-\Big) dm. \tag{2.10}$$

At $m = m_*$, $S = 2m_*$, the operator in parentheses has norm bounded below by 1/2 on one-forms, by the reverse triangle inequality. Thus normalization of the Hopf form does not cancel this radial derivative. The actual projection errors must also be controlled; (2.10) is not substituted as an exact identity for the approximate sequence.

2.2 What the next differentiated equation must control

Weak control of the length derivative is available without resolving $\mathscr{B}$. The exact Clifford identity

$$\nu := 2\rho^2|\Pi|^2 - m = 2\rho|\zeta|^2 + |c(\zeta)\Phi|^2$$

gives $\fint \nu \leqslant C\eta$. Cauchy–Schwarz in (1.2), together with the bounded average of S, gives $\fint |m - m_*| \leqslant C\eta^{1/4}$. More precisely, for bounded Lipschitz F, splitting into $S \leqslant L$ and its complement, with a fixed sufficiently large L, gives

$$\fint |F(m) - F(m_*)|^2 \leqslant C_F \sqrt{\eta}. \tag{2.11}$$

On the first set use (1.2) and $\nu^2 \leqslant C_L \nu$; on the second use boundedness of F and $(S - 2m_*)^2/S \geqslant cS$. Consequently $dF(m)$ is small as a distribution tested against one-forms with bounded normalized L^2 divergence, with bound $C_F \eta^{1/4}$. A cutoff depending only on ρ preserves this conclusion by $d\rho$ being small. A cutoff in S does not follow at this strength: its derivative contains H, and the extra product is only L^1-small against bounded tests.

This weak statement does not control the derivative of the normalized phase connection $\theta_\tau = j_\tau/m$. The next Ricci interchange in (2.10) contains $d(J_+ d\log m)$, not just dm in a negative norm. The distinction can be measured on the already available local actual simple-zero solutions. Their rescaled limit on a fixed mass shell is

$$\tau = (\bar z_1, \bar z_2)/\ell, \qquad m = |x|^2/\ell^2, \qquad 0 < m_1 < m < m_2.$$

Direct polar integration gives

$$\int |d\theta_\tau|^2 = 4\pi^2 \log(m_2/m_1), \qquad \int \frac{|dm|^2}{m^2} |\nabla(\tau/|\tau|)|^2 = 12\pi^2 \log(m_2/m_1). \tag{2.12}$$

Indeed $|d\theta_\tau|^2 = 4/r^4$, $|dm|^2/m^2 = 4/r^2$, $|\nabla(\tau/|\tau|)|^2 = 3/r^2$, and $\mathrm{Area}(S^3) = 2\pi^2$. For a radial cutoff decreasing from one to zero,

$$-\frac{1}{4\pi^2} \int d\chi \wedge \theta_\tau \wedge d\theta_\tau = 1. \tag{2.13}$$

The actual local corrections converge in the rescaled differentiable norms on this nonzero shell, so (2.12) persists up to $o(1)$; the corresponding local flux is $1 + o(1)$. Meanwhile the shell's phase L^2 norm and bounded truncation's negative derivative norm tend to zero. The weighted controls in Sections 1–2 and the covariance contribution on each such shrinking core can also tend to zero, as in I51's scale calculation.

Thus the phase curvature is a specific remaining critical quantity. The calculation does not furnish a closed calibrated zero-end sequence: it measures what the local equations permit before matching to the exterior. A global cancellation would have to be proved using that matching or further actual equations, not negative-norm convergence alone.

3. The actual shrinking-surgery class, with arbitrary backgrounds

Let X be a closed complex-hyperbolic surface, $A = c_1^2(X) > 0$, and fix

$$M = X\#k\overline{\mathbb{CP}}^2\#\ell(S^1 \times S^3), \quad Q = k + 4\ell = Ah, \quad C_Q = 8\pi^2 Q, \quad 0 < h \leqslant 1/3.$$

Here Q is the surgery number, unrelated to the ASD form in Section 1. Retain the actual polarized scalar budget of this family, for X and M. On a metric g on X put

$$P_g = \int |\operatorname{Ric}^\circ|^2, \quad S_g = \int s^2, \quad R_g = P_g + S_g/4, \quad L_g = S_g - (\int s)^2/V_g.$$

By *standard shrinking surgery* we mean inserting scalar-flat ASD Burns pieces at blowups and conformally flat scalar-flat Schwarzschild necks at the handles, with the unchanged exterior of the fixed smooth metric g. This is the classical construction used in LeBrun, Theorem 4.3, applied in normal-coordinate balls. It is not an arbitrary metric on M. The precise limits are

$$V_\epsilon \to V_g, \quad \int s_\epsilon \to \int s_g, \quad S_\epsilon \to S_g, \quad \int |W_\epsilon^+|^2 \to \int |W_g^+|^2, \quad P_\epsilon \to P_g + C_Q, \quad R_\epsilon \to R_g + C_Q. \tag{3.1}$$

The last two follow from Chern–Weil and $T_+(M) = T_+(X) - Q$. One can arrange uniformly bounded negative scalar curvature at fixed g, k, ℓ. At cutting radius R, the background normal-coordinate cutoff has bounded curvature; an ALE scale $a = o(R^2)$ makes the model contribution $O(a^2/R^4)$ bounded on the transition annulus. The interiors are scalar-flat, and the changed volume tends to zero. All constants may depend on this fixed background and these finite surgery data. For a varying family the radii are subsequently chosen diagonally, not by an unjustified uniform injectivity estimate.

The raw limiting relative Einstein defect is therefore

$$\mathscr{F}_Q(g) = 4 - \frac{(\int s_g)^2}{V_g(R_g + C_Q)}. \tag{3.2}$$

3.1 Conformal minimization before and after surgery

Let $\widehat{g}$ be the scalar-minus-four Yamabe representative of $[g]$, and put $\delta = S_g - S_{\widehat{g}} \geqslant 0$. I41 gives $P_g = P_{\widehat{g}} + \delta/12$ and $R_g = R_{\widehat{g}} + \delta/3$. Define

$$\alpha_Q = \frac{4(P_{\widehat{g}} + C_Q)}{R_{\widehat{g}} + C_Q}.$$

Direct substitution proves the exact identity

$$\boxed{\mathscr{F}_Q(g) - \alpha_Q = \frac{(1-\alpha_Q)\delta/3 + L_g}{R_g + C_Q}.} \tag{3.3}$$

If $\alpha_Q < 1$, Yamabe normalization minimizes this raw limiting functional in the conformal class. If $\alpha_Q \geqslant 1$, use instead

$$\mathscr{F}_Q(g) = \frac{\alpha_Q(R_{\widehat{g}} + C_Q) + \delta/3 + L_g}{R_{\widehat{g}} + C_Q + \delta/3} \geqslant 1.$$

Normalization **after** surgery needs a separate argument; (3.3) alone does not identify its limit for a nonconstant-scalar background.

Lemma 3 (negative Yamabe continuity for these shrinking surgeries). For fixed g, k, ℓ, the above construction can be chosen so that

$$Y(M, [g_\epsilon]) \longrightarrow Y(X, [g]) < 0. \tag{3.4}$$

Proof. Cut off the background Yamabe minimizer on the shrinking coordinate balls. Their total Dirichlet capacity tends to zero, which gives the upper bound on the limit in (3.4). In particular the surgery Yamabe constants are bounded above by a fixed negative number for small scales. Normalize their positive minimizers by $\int u_\epsilon^4 = 1$. The uniform bound $s_\epsilon^- \leqslant K$, the equation

$$6\Delta u_\epsilon + s_\epsilon u_\epsilon = Y_\epsilon u_\epsilon^3,$$

and the maximum principle give $\|u_\epsilon\|_\infty^2 \leqslant K/|Y_\epsilon| \leqslant C$. The energy equation also bounds $\int |du_\epsilon|^2$: explicitly $Y_\epsilon \geqslant -KV_\epsilon^{1/2}$ and $6\int |du_\epsilon|^2 \leqslant KV_\epsilon^{1/2}$. Choose a cutoff χ_ϵ which vanishes on all changed cores, is one outside slightly larger unchanged coordinate balls, and has $\int |d\chi_\epsilon|^2 \to 0$. Extend $\chi_\epsilon u_\epsilon$ by zero to X. The removed L^4 mass tends to zero by the volume and supremum bounds. On expanding its Yamabe numerator, discard the removed nonnegative gradient term. The removed negative scalar term tends to zero, and the cutoff cross term is bounded by

$$C\|u_\epsilon\|_\infty \|d\chi_\epsilon\|_2 \|du_\epsilon\|_2 = o(1).$$

Thus $Y(X, [g]) \leqslant Y_\epsilon + o(1)$, proving the lower bound. QED.

Conformal invariance of Weyl energy and I41's exact conformal identity now imply, for scalar-minus-four representatives of the outputs,

$$\widehat{V}_\epsilon \to V_{\widehat{g}}, \qquad \widehat{P}_\epsilon \to P_{\widehat{g}} + C_Q, \qquad \widehat{R}_\epsilon \to R_{\widehat{g}} + C_Q, \qquad \mathscr{E}(\widehat{g}_\epsilon) \to \alpha_Q. \tag{3.5}$$

3.2 Exact restricted infimum and the first-order penalty

Proposition 4. The infimum of the limiting defects in (3.2) and (3.5), over all smooth backgrounds on X, is

$$\boxed{\frac{4h}{1+h}.} \tag{3.6}$$

This is an infimum in the specified construction class, not an infimum over all metrics on the balanced M.

Proof. On the Yamabe background the scalar budget and the opposite chiral identity on X, where $T_- = A/3$, give

$$2\pi^2 A \leqslant V_{\widehat{g}} \leqslant 2\pi^2 A + 3P_{\widehat{g}}/4.$$

Consequently

$$\alpha_Q \geqslant \frac{4(P_{\widehat{g}} + 8\pi^2 Ah)}{8\pi^2 A(1+h) + 4P_{\widehat{g}}} \geqslant \frac{4h}{1+h}. \tag{3.7}$$

For the raw functional use the two cases after (3.3). The complex-hyperbolic background gives equality in the limiting value. For $h < 1/3$ the quantitative excess in (3.7) is

$$\alpha_Q - \frac{4h}{1+h} \geqslant \frac{4(1-3h)P_{\widehat{g}}}{(1+h)\{8\pi^2 A(1+h) + 4P_{\widehat{g}}\}}. \tag{3.8}$$

There is a stronger exact way to see the classical content. Write $B_g = \int_X |W_g^-|^2$. The opposite Chern–Weil identity gives $R_g = 8\pi^2 A + 4P_g - 12B_g$, and hence

$$\boxed{\mathscr{F}_Q(g) - \frac{4h}{1+h} = \frac{4(1-3h)P_g + 48hB_g + (1+h)L_g}{(1+h)(R_g + C_Q)}.} \tag{3.9a}$$

Thus the finite-h lower bound does not even require the scalar monopole budget. For $0 < h < 1/3$, equality in this limit forces $P_g = B_g = L_g = 0$. At $h = 1/3$ it only forces $B_g = L_g = 0$. Eliminating P_g gives the equivalent form

$$\mathscr{F}_Q(g) - \frac{4h}{1+h} = \frac{(1-3h)(S_g - 32\pi^2 A)/3 + 16\int_X |W_g^-|^2 + (1+h)L_g}{(1+h)(R_g + C_Q)}. \tag{3.9b}$$

This is the same scalar–signature mechanism as I44, not a new independent differential obstruction. The new use here keeps arbitrary backgrounds and the post-surgery conformal limit explicit. QED.

For $h_j \to 0$, take all surgery errors to be $o(Q_j)$, after fixing each background. If the normalized outputs have defect $O(h_j)$, (3.8) forces $P_{\widehat{g}_j} = O(Q_j)$, and

$$\boxed{\frac{\mathscr{E}(\widehat{g}_{j,\epsilon_j})}{h_j} = 4 + \frac{P_{\widehat{g}_j}}{2\pi^2 Q_j} + o(1).} \tag{3.10}$$

For competitive **raw** outputs, (3.3) additionally forces $\delta_j, L_{g_j} = O(Q_j)$, and

$$\frac{\mathscr{E}(g_{j,\epsilon_j})}{h_j} = 4 + \frac{P_{\widehat{g}_j}}{2\pi^2 Q_j} + \frac{\delta_j}{24\pi^2 Q_j} + \frac{L_{g_j}}{8\pi^2 Q_j} + o(1). \tag{3.11}$$

In this class, leading coefficient four is equivalent to the displayed nonnegative penalties being $o(Q_j)$. In (3.10) only the first penalty remains. A claim that the two extra penalties survive normalization on M would be incorrect. No uniform geometry across j was used: the fixed-j limits (3.1) and (3.4) allow a diagonal with every stated error $o(Q_j)$.

4. What a perturbation of these metrics must change

Let g_j^0 be a complex-hyperbolic shrinking-surgery sequence chosen with scalar error $\int (s+4)^2 = o(Q_j)$. Let C_j contain all changed cores and transition regions, and $N_j = M_j \setminus C_j$ be the exact complex-hyperbolic exterior. Then

$$\int_{C_j} |\operatorname{Ric}^{\circ}_{g_j^0}|^2 = 8\pi^2 Q_j + o(Q_j).$$

Consider actual positive metrics $g_j = g_j^0 + \epsilon_j q_j$, with $\epsilon_j = O(\sqrt{h_j})$, and identify tensor L^2 spaces by pointwise isometries and the volume density. Assume uniform small metric comparison, and the following explicit expansion, rather than an unproved uniform differentiability on collapsing cores:

$$T_j = T_j^0 + \epsilon_j L_j q_j + e_j, \quad \|e_j\|_{2,C_j} = o(\sqrt{Q_j}), \quad \epsilon_j^2 \|L_j q_j\|_{2,C_j}^2 = o(Q_j). \tag{4.1}$$

Here T_j is the identified traceless Ricci tensor and L_j its identified linearization. The analogous exterior expansion is assumed with $o(\sqrt{Q_j})$ remainder and $\epsilon_j \|L_j q_j\|_{2,N_j} = O(\sqrt{Q_j})$. The cross term on the cores is $o(Q_j)$ by Cauchy–Schwarz; the exterior has $T_j^0 = 0$. Therefore

$$P_{g_j} = 8\pi^2 Q_j + \epsilon_j^2 \|L_j q_j\|_{2,N_j}^2 + o(Q_j). \tag{4.2}$$

If also $\int (s_{g_j} + 4)^2 = o(Q_j)$, I51's exact scalar comparison shows that Yamabe normalization changes P by $o(Q_j)$. Competitive normalized outputs on the balanced family then satisfy

$$\frac{\mathscr{E}(\widehat{g}_j)}{h_j} = 4 + \frac{\epsilon_j^2 \|L_j q_j\|_{2,N_j}^2}{2\pi^2 Q_j} + o(1). \tag{4.3}$$

TT variations with uniformly controlled physical derivatives and integrated core linearization energy $O(Q_j)$ satisfy the smallness in (4.1). The exterior first scalar variation vanishes because it is Einstein

and the variation is TT. Control in an expanding rescaled ALE domain must be **integrated**; a rescaled C^2 supremum alone does not supply (4.1).

Conversely retain the $o(\sqrt{Q_j})$ expansion remainder but not the core energy smallness. Suppose the normalized sequence has $\mathscr{E}/h_j \leqslant 4-\sigma+o(1)$, for $0<\sigma<4$, and retain the scalar-error hypothesis. The balanced chiral and scalar budgets give $\widehat{V}_j = 2\pi^2 A_j + O(Q_j)$, whence

$$P_{g_j} \leqslant 8\pi^2 Q_j(1-\sigma/4) + o(Q_j).$$

The reverse triangle inequality on C_j forces

$$\boxed{\liminf_j \frac{|\epsilon_j|\|L_j q_j\|_{2,C_j}}{\sqrt{Q_j}} \geqslant \sqrt{8\pi^2}\left(1-\sqrt{1-\sigma/4}\right) > 0.} \tag{4.4}$$

Thus at amplitude $O(\sqrt{h})$, a successful perturbation with a controlled expansion needs core linearized energy of order at least A, not merely Ah. This is a necessary escape scale within the specified perturbative description. It neither constructs that escape nor excludes a genuinely nonperturbative geometry. Arbitrarily changing an isolated core while retaining cheap inverse-surgery collars is already covered by I44's Weyl-only cap inequality, and is not counted again here.

There is also a necessary sign. The upper energy bound and (4.1) first give $|\epsilon_j|\|L_j q_j\|_{2,C_j} = O(\sqrt{Q_j})$, so squaring the expansion is legitimate. Any proposed improvement as above must obey

$$-\epsilon_j \langle T_j^0, L_j q_j\rangle_{L^2(C_j)} \geqslant \pi^2\sigma Q_j + \tfrac{1}{2}\epsilon_j^2 \|L_j q_j\|_{2,C_j}^2 + o(Q_j). \tag{4.5}$$

Large core derivative energy alone is not enough: it must cancel the existing core Ricci tensor with this sign. This statement uses the raw tensor expansion and the scalar identity $P_{g_j} = P_{\widehat{g}_j} + o(Q_j)$. It does **not** infer tensor closeness from that scalar energy identity.

4.1 A conformal tensor estimate without a Sobolev constant

The preceding distinction can be resolved under a stronger, explicit scalar hypothesis. This auxiliary lemma does not require gauge theory.

Lemma 5. On a closed four-manifold let $\widehat{g} = e^{2f} g$ have scalar curvature -4, and write $Z = \mathrm{Ric}^\circ_g$, $\widehat{Z} = \mathrm{Ric}^\circ_{\widehat{g}}$ as covariant two-tensors. If $\eta_f = \|f\|_\infty$ is sufficiently small, then

$$\boxed{\|\widehat{Z} - Z\|_{2,g} \leqslant C\{\|s_g + 4\|_{2,g} + \eta_f \|Z\|_{2,g}\}.} \tag{4.6}$$

The constant is dimensional. No pointwise Ricci, diameter, or Sobolev bound is used.

Proof. All norms in this proof are for g, and $\Delta = d^*d$. Put

$$B = \widehat{Z} - Z = -2(\nabla^2 f)_0 + 2(df\otimes df)_0, \quad T = \|B\|_2, \quad H = \|\nabla^2 f\|_2, \quad D = \|\Delta f\|_2, \quad G = \|df\|_4^2, \quad a = \|s_g + 4\|_2.$$

Conformal invariance of the L^2 norm of covariant two-tensors in dimension four and $\operatorname{div} Z = ds/4$ give the exact formula

$$T^2 = P_{\widehat{g}} - P_g - \int (s_g + 4)\Delta f - 4\int Z(df, df). \tag{4.7}$$

The negative constant-scalar representative is the Yamabe minimizer. The conformal Chern–Weil identity therefore gives $P_{\widehat{g}} - P_g \leqslant 0$, and

$$T^2 \leqslant aD + 4\sqrt{P_g}\, G. \tag{4.8}$$

Integrating $\operatorname{div}(f|df|^2\nabla f)$ proves $G \leqslant \eta_f(2H + D) \leqslant 4\eta_f H$. The apparent volume term in the scalar equation is unnecessary. Set $u = e^f$, $v = u - 1$, $e = s_g + 4$. The exact equation is

$$6\Delta v + 4u(u + 1)v = -eu.$$

Testing with v, and using the uniform upper and positive lower bounds on u, gives $\|v\|_2 + \|dv\|_2 \leqslant Ca$. In particular $\|e^{2f} - 1\|_2 \leqslant Ca$. Trace splitting and the conformal scalar equation now give

$$H \leqslant T/2 + G + D/2, \qquad 6\Delta f = -s_g - 4e^{2f} + 6|df|^2, \qquad D \leqslant Ca + G.$$

For sufficiently small η_f, absorption yields

$$H \leqslant C(T + a), \quad G \leqslant C\eta_f(T + a), \quad D \leqslant Ca + C\eta_f T.$$

Substitute into (4.8) and solve the resulting quadratic inequality in T. This proves (4.6), and also

$$\|\nabla^2 f\|_2 + \|\Delta f\|_2 \leqslant C(a + \eta_f\sqrt{P_g}), \qquad \|f\|_2 + \|df\|_2 \leqslant Ca.$$

QED.

In particular, if

$$|s_g + 4|_\infty = O(h), \quad P_g = O(Ah), \quad V_g = O(A),$$

then the maximum principle in the Yamabe equation gives $\eta_f = O(h)$. Formula (4.6) implies

$$\|\widehat{Z} - Z\|_{2,g} = O(h\sqrt{A}) = o(\sqrt{Ah}). \tag{4.9}$$

This implication includes the extra pointwise scalar hypothesis; $\int (s_g + 4)^2 = o(Ah)$ alone was not used to claim (4.9).

4.2 Nonzero diffuse TT variations at the first-order scale

The class in (4.2)–(4.3) is not restricted to a formal or zero variation. On the same covering-and-surgery family there are actual metrics, after Yamabe normalization, with

$$\boxed{\frac{\mathscr{E}(\widehat{g}_j)}{h_j} = 4 + \frac{\tau^2 c_0}{2\pi^2} + o(1), \qquad c_0 > 0,} \tag{4.10}$$

for each fixed real τ. Here is a construction and its quantifiers.

Reserve one small evenly covered chart in the fixed complex-hyperbolic base, and all its lifts in X_j, before choosing surgery points. All finitely many surgery points can be chosen in the complement. There are $d_j = A_j/A_{\text{base}}$ disjoint reserved charts. Let $b_{j,t}$ be the raw shrinking-surgery metric and

$g^0_{j,t} = u^2_{j,t} b_{j,t}$ its scalar-minus-four representative. For each fixed j, Lemma 3, the uniform supremum estimate in its proof, elliptic bootstrapping away from the punctures, and removability of bounded solutions imply $u_{j,t} \to 1$ smoothly on all reserved compact charts. The limit is one by uniqueness of the negative constant-scalar representative. Choose t_j so that the error on these finitely many charts is $o(h_j^2)$ in C^8, simultaneously with all earlier energy errors. This is a diagonal choice, not a uniform injectivity assertion.

On these charts g_j^0 is exactly conformal to the complex-hyperbolic metric, so it is Bach-flat. Let $\mathscr{B}_{\mathrm{Bach}}$ denote the Bach tensor and let a_j be the sum of lifted compactly supported symmetric seed tensors. Set

$$q_j = D\mathscr{B}_{\mathrm{Bach}}|_{g_j^0}(a_j).$$

This is a smooth global TT tensor, extended by zero: differentiate the trace and divergence identities for the Bach tensor, noting that the background Bach tensor vanishes on the support. These identities and the transverse-traceless principal symbol are classical; see Gursky–Viaclovsky, Section 1.4 and Theorem 3.10. The order-six principal symbol of $D\,\mathrm{Ric}^{\circ} \circ D\mathscr{B}_{\mathrm{Bach}}$ is a nonzero multiple of $|\xi|^6$ on transverse trace-free symbol tensors. For an explicit seed, take normal coordinates at a point, a nonzero symmetric matrix H_0 with $(H_0)_{1i} = 0$, $\operatorname{tr} H_0 = 0$, and $a = \chi(x)(x^1)^6 H_0$, where χ is compactly supported and equals one near that point. Every jet below order six vanishes there; the displayed principal symbol makes the limiting linearized Ricci tensor nonzero. Thus q_j has uniformly bounded physical C^2 norm and

$$\int |L_j q_j|^2 = (c_0 + o(1))A_j, \qquad c_0 > 0.$$

The positive metrics $g_j = g_j^0 + \tau\sqrt{h_j}\, q_j$ are unchanged near every core. Their first scalar variation is $-\langle Z_j^0, q_j\rangle = o(h_j^2)$ on the support, and their second variation is uniformly bounded. Hence $|s_{g_j} + 4|_\infty = O(h_j)$. The Ricci Taylor remainder has L^2 norm $O(h_j\sqrt{A_j}) = o(\sqrt{Q_j})$, and

$$P_{g_j} = 8\pi^2 Q_j + \tau^2 c_0 Q_j + o(Q_j).$$

Lemma 5 controls the final conformal tensor change by $o(\sqrt{Q_j})$; the exact scalar energy identity also gives $P_{\widehat{g}_j} - P_{g_j} = o(Q_j)$. The balanced-family scalar and chiral budgets give $\widehat{V}_j = 2\pi^2 A_j + O(Q_j)$, proving (4.10). Actual monopole supply persists by the same all-metric topological input as before; no locally constructed spinor replaces that input.

This construction measures a genuine diffuse first-order payment. It does not establish positivity in every reduced TT direction, exclude core-correlated modes satisfying (4.5), or determine the unrestricted optimal coefficient.

4.3 First-order stability on almost-critical recovery sequences

For suitably chosen recovery metrics the perturbative exclusion is stronger than the core norm threshold (4.4). It covers core-supported directions as well, provided the full curvature expansion is valid at the required scale.

Choose the normalized shrinking-surgery bases g_j^0 so that

$$P_j^0 = 8\pi^2 Q_j + o(Q_j), \quad V_j^0 = O(A_j), \quad \mathrm{Vol}(C_j) = o(Q_j), \quad P^0_{N_j} = o(Q_j^2/A_j). \tag{4.11}$$

These choices are compatible with Section 4.2. For each fixed j, first choose small exclusion balls about its finitely many surgery centers, then shrink the gluing scales further. Lemma 3 and local smooth convergence give the exterior bound; the uniform supremum bound on the conformal factor gives the core volume bound.

Let $\mathscr{B}_W = \operatorname{grad} \int |W|^2$, in the operator Weyl convention used throughout the project. On the raw construction this gradient vanishes on the unchanged Einstein exterior and on the scalar-flat ASD or conformally flat donors. On each gluing annulus of radius R_i, the rescaled metric is C^4-close to Euclidean with error $\varepsilon_i = O(R_i^2 + (a_i/R_i)^2)$. This fourth-order assertion follows from the explicit donors: the Burns potential $r^2 + a_i^2 \log r^2$ and the Schwarzschild metric $(1 + a_i^2/r^2)^2 g_{\mathrm{Eucl}}$ have

$$|\partial^k (g - g_{\mathrm{Eucl}})| \leqslant C_k a_i^2 r^{-2-k}, \qquad 0 \leqslant k \leqslant 4$$

on either relevant asymptotically flat chart. Fixed rescaled smooth cutoffs preserve these bounds. A second-derivative gluing estimate alone would not have justified the following Bach estimate.

Fourth-order locality now gives $\int_{\mathrm{annulus}_i} |\mathscr{B}_W| \leqslant C\varepsilon_i$. There are $O(Q_j)$ annuli and their errors can tend uniformly to zero. The covariant Bach gradient has conformal weight -2; its L^1 norm in four dimensions is conformally invariant. Therefore

$$\|\mathscr{B}_W(g_j^0)\|_1 = o(Q_j), \qquad \|Z_j^0\|_1 \leqslant \sqrt{P_{C_j}^0 \operatorname{Vol}(C_j)} + \sqrt{P_{N_j}^0 V_j^0} = o(Q_j). \tag{4.12}$$

From $P = 2\int |W|^2 + S/12 - 16\pi^2\chi$ and $s = -4$,

$$\operatorname{grad} P = 2\mathscr{B}_W + \tfrac{2}{3} Z, \qquad \|\operatorname{grad} P(g_j^0)\|_1 = o(Q_j). \tag{4.13}$$

Proposition 6. Let q_j be smooth symmetric tensors with $\|q_j\|_\infty \leqslant C$. Normalize the positive path $g_j^0 + tq_j$ to scalar -4, denoting it by $\widehat{g}_j(t)$. Take $t_j = O(\sqrt{h_j})$. Identify the final traceless Ricci tensors isometrically with the fixed L^2 space at g_j^0, including the square root of the volume density, and write

$$\mathscr{Z}_j(t_j) = Z_j^0 + t_j \dot{\mathscr{Z}}_j(0) + e_j. \tag{4.14}$$

If $\|e_j\|_2 = o(\sqrt{Q_j})$ and the outputs are competitive, $\mathscr{E}(\widehat{g}_j(t_j)) = O(h_j)$, then

$$P_{\widehat{g}_j(t_j)} = 8\pi^2 Q_j + t_j^2 \|\dot{\mathscr{Z}}_j(0)\|_2^2 + o(Q_j), \qquad \liminf_j \frac{\mathscr{E}(\widehat{g}_j(t_j))}{h_j} \geqslant 4. \tag{4.15}$$

No TT hypothesis is required for this conditional assertion.

Proof. The identification is essential. For initial velocity $v = \dot{g}$, $H_0 = g^{-1}v$, its derivative is

$$\dot{\mathscr{Z}} = \dot{Z} - \tfrac{1}{2}(H_0 Z + Z H_0) + \tfrac{1}{4}(\operatorname{tr} H_0) Z, \qquad 2\langle Z, \dot{\mathscr{Z}} \rangle_{L^2} = P'(v). \tag{4.16}$$

For a conformal velocity $2\phi g$, the identification corrections cancel in four dimensions, and $P'(2\phi g) = 0$ at constant scalar curvature. Hence the normalized path derivative satisfies

$$2\langle Z_j^0, \dot{\mathscr{Z}}_j(0) \rangle = P'_{g_j^0}(q_j) = o(Q_j)$$

by (4.13). The scalar and chiral budgets turn competitiveness into $P_{\widehat{g}_j(t_j)} = O(Q_j)$ and $\widehat{V}_j = 2\pi^2 A_j + O(Q_j)$. The triangle inequality in (4.14) gives $|t_j| \|\dot{\mathscr{Z}}_j(0)\|_2 = O(\sqrt{Q_j})$. Squaring (4.14), with its stated remainder, proves (4.15). QED.

There is a quantitative converse that does not assume a small remainder. Define e_j by the exact difference (4.14). If an actual normalized path with the same bounded initial direction satisfies $\mathscr{E}/h_j \leqslant 4-\sigma+o(1)$, $0<\sigma<4$, then

$$\boxed{\liminf_j \frac{\|e_j\|_2^2}{Q_j} \geqslant 8\pi^2\left(1-\sqrt{1-\sigma/4}\right)^2.} \tag{4.17}$$

Indeed (4.13) implies $\|Z_j^0+t_j\dot{\mathscr{Z}}_j(0)\|_2^2 \geqslant (8\pi^2+o(1))Q_j$, whereas the final squared norm is at most $(8\pi^2(1-\sigma/4)+o(1))Q_j$. Apply the reverse triangle inequality. This bound is a Hilbert-space consequence with an actual geometric hypothesis; it does not construct such a path.

Thus on these recovery bases, simply arranging a large core linearization cannot satisfy (4.5) within the small-remainder class. A putative improvement must create the quantitatively nonnegligible nonlinear curvature remainder (4.17), or use a different base geometry. Neither possibility has been excluded.

4.4 Competitive energy supplies the missing Taylor control

The remainder hypothesis in Proposition 6 can be removed in a substantially larger, explicitly gauged class. In this section all derivatives and unmarked norms are for g_j^0.

Theorem 7 (small-amplitude TT competitors). Let g_j^0 be the recovery sequence of (4.11)–(4.13). Let k_j be any smooth symmetric tensors with

$$\operatorname{tr}_{g_j^0} k_j = 0, \qquad \delta_{g_j^0} k_j = 0, \qquad a_j = \|k_j\|_\infty \longrightarrow 0.$$

Suppose k_j is supported in U_j, and put $K_j = \|\operatorname{Rm}(g_j^0)\|_{L^2(U_j)}$. If

$$a_j^2 K_j = o(\sqrt{Q_j}), \qquad \mathscr{E}(\widehat{g_j^0+k_j}) = O(h_j), \tag{4.18}$$

where the hat denotes negative Yamabe normalization, then

$$\boxed{\liminf_j \frac{\mathscr{E}(\widehat{g_j^0+k_j})}{h_j} \geqslant 4.} \tag{4.19}$$

There are no assumed derivative bounds or Taylor remainders. Globally $K_j = O(\sqrt{A_j})$: for the full Riemann tensor norm, $\int |\operatorname{Rm}|^2 = 32\pi^2\chi + 4P$, and $\chi(M_j) = O(A_j)$. Thus $a_j = o(h_j^{1/4})$ is sufficient.

Proof. We first record the geometry-independent tensor estimate that supplies the second derivatives. On any closed manifold, for a smooth tensor k, put $a = \|k\|_\infty$ and $H = \|\nabla^2 k\|_2$. Integration by parts gives

$$\|\nabla k\|_4^2 \leqslant CaH. \tag{4.20}$$

To see this, integrate the divergence of the vector field with components $|\nabla k|^2\langle k, \nabla_i k\rangle$, and bound the resulting Hessian terms by $CaH\|\nabla k\|_4^2$. For the Hessian estimate regard $\alpha = \nabla k$ as a one-form valued in $E = S^2T^*M$. The bundle-valued Weitzenbock identity gives

$$H^2 = \|d^\nabla \alpha\|_2^2 + \|\delta^\nabla \alpha\|_2^2 - \int \langle \mathscr{R}\alpha, \alpha\rangle.$$

Here $d^{\nabla}\alpha = F_E k$, $\delta^{\nabla}\alpha = \nabla^*\nabla k$, and $|F_E| + |\mathscr{R}| \leqslant C|\operatorname{Rm}|$. Consequently

$$H^2 \leqslant \|\nabla^*\nabla k\|_2^2 + Ca^2K^2 + CK\|\nabla k\|_4^2 \leqslant \|\nabla^*\nabla k\|_2^2 + Ca^2K^2 + CaKH. \tag{4.21}$$

This classical identity is stated, for example, in Oh–Kim, Appendix 23. No curvature derivative is taken. Curvature is only integrated where k or its derivatives is supported, with no artificial boundary. Equations (4.20)–(4.21) imply

$$H \leqslant C\{\|\nabla^*\nabla k\|_2 + aK\}.$$

For an exactly TT tensor the principal part of $DZ_{g^0}(k)$ is one half of this rough Laplacian; every other term is $\operatorname{Rm} * k$. Therefore

$$H \leqslant C\{\|DZ_{g^0}(k)\|_2 + aK\}. \tag{4.22}$$

Neither (4.20) nor (4.21) uses a Sobolev or injectivity constant.

For a sufficiently small, the connection-difference formula for $g = g^0 + k$ gives exact Ricci and scalar Taylor remainders bounded by

$$C\{aH + \|\nabla k\|_4^2 + a^2K\} \leqslant C(aH + a^2K). \tag{4.23}$$

This follows by expanding the inverse metric once: the second-order terms are bounded by $C(a|\nabla^2 k| + |\nabla k|^2 + a^2|\operatorname{Rm}|)$. Here and below the constants are uniform under a small. The first scalar variation in TT gauge is $-\langle Z^0, k\rangle$. With $b = \|s_g + 4\|_{2,g}$ and $\|Z^0\|_2 = O(\sqrt{Q})$,

$$b \leqslant C(a\sqrt{Q} + aH + a^2K). \tag{4.24}$$

The competitive energy is assumed for the final Yamabe metric, not silently transferred to g. The balanced-family scalar and chiral budgets give $P_{\hat{g}} = O(Q)$. The exact conformal identity and the negative Yamabe variational inequality give

$$0 \leqslant P_g - P_{\hat{g}} = \frac{S_g - Y([g])^2}{12} \leqslant \frac{b^2}{12}. \tag{4.25}$$

For completeness, if $I = \int s_g < 0$, then $Y([g])^2 \geqslant I^2/V_g$, and $S_g - I^2/V_g \leqslant \int (s_g + 4)^2$. If $I \geqslant 0$, the latter integral is at least S_g, which also proves (4.25). The actual family supplies $Y([g]) < 0$. Thus $\sqrt{P_g} \leqslant C\sqrt{Q} + Cb$, without a bound on the normalizing conformal factor.

Insert the nonlinear Ricci expansion into (4.22), use (4.23)–(4.25), and absorb the terms CaH. This gives

$$H \leqslant C(\sqrt{Q} + aK), \qquad b + \|\operatorname{Rem}_Z\|_2 \leqslant C(a\sqrt{Q} + a^2K). \tag{4.26}$$

By (4.18), both terms on the right of the second inequality are $o(\sqrt{Q})$. In particular $P_g - P_{\hat{g}} = o(Q)$.

It remains to preserve the metric and volume factors in the tensor expansion. Let I_k be the norm-preserving identification used in (4.16), and $\dot{\mathscr{Z}}_0(k)$ its derivative. Smooth pointwise matrix calculus, (4.23), and (4.26) give

$$\|I_k Z_g - Z^0 - \dot{\mathscr{Z}}_0(k)\|_2 \leqslant C(a\sqrt{Q} + a^2K) = o(\sqrt{Q}). \tag{4.27}$$

The extra terms are bounded by $Ca\|DZ(k)\|_2 + Ca^2\|Z^0\|_2$ in addition to (4.23), so no uncontrolled change of volume has been suppressed. Since $P_g = O(Q)$, the linearized tensor also has norm $O(\sqrt{Q})$. Consequently

$$P_g = P_{g^0} + P'_{g^0}(k) + \|\dot{\mathscr{Z}}_0(k)\|_2^2 + o(Q) \geqslant 8\pi^2 Q + o(Q). \tag{4.28}$$

Here $|P'_{g^0}(k)| \leqslant a\|\operatorname{grad} P(g^0)\|_1 = o(Q)$. Combine with (4.25)–(4.26) and $\widehat{V} = 2\pi^2 A + O(Q)$ to obtain (4.19). QED.

On the chosen surgery cores, $\|\operatorname{Rm}(g_j^0)\|^2_{L^2(C_j)} = O(Q_j)$: the Weyl energy is the conformally invariant donor-and-annulus energy, the Ricci energy is $O(Q_j)$, and the scalar term is controlled by the core volume. Thus every vanishing-amplitude core-supported TT competitor satisfies (4.18), even with arbitrarily high derivatives. For perturbations leaving the old cheap collars unchanged, I44's cap argument already gives the exclusion; this special case is a consistency check, not a new application. The broader assertion here allows global TT changes with no collar derivative bound.

There is a related gauge formulation without a TT decomposition.

Corollary 8 (constant-scalar Bianchi slice). Suppose instead that the actual endpoints $g_j = g_j^0 + k_j$ already satisfy

$$s_{g_j} = -4, \quad \beta_{g_j^0} k_j := \delta_{g_j^0} k_j + \tfrac{1}{2} d \operatorname{tr}_{g_j^0} k_j = 0, \quad \|k_j\|_\infty = o(h_j^{1/4}), \quad \mathscr{E}(g_j) = O(h_j). \tag{4.29}$$

Then $\liminf \mathscr{E}(g_j)/h_j \geqslant 4$.

Proof. Use $F(g) = \operatorname{Ric}_g + g$. In this gauge $DF(k)$ has principal part $\frac{1}{2}\nabla^*\nabla k$; the remaining terms are $\operatorname{Rm} * k + k$. Replace K in (4.22) by $\widetilde{K} = K + \sqrt{V_{g^0}} = O(\sqrt{A})$. At both endpoints $F = Z$, of norm $O(\sqrt{Q})$. The same connection-difference formula and absorption give

$$H \leqslant C(\sqrt{Q} + a\widetilde{K}), \qquad \|\operatorname{Rem}_F\|_2 \leqslant C(a\sqrt{Q} + a^2\widetilde{K}) = o(\sqrt{Q}).$$

Finally use $\int |F|^2 = P + \frac{1}{4}\int (s+4)^2$. This functional equals P at both endpoints and its first derivative equals P' at g^0. The identified expansion therefore has exactly the sign argument in (4.28). QED.

These are actual, derivative-free small-amplitude assertions in the specified gauges, not assertions that arbitrary nearby metrics admit a uniformly controlled gauge change. The radius $o(h^{1/4})$ is sufficient; its sharpness is not proved. A successful sub-four construction must lie outside these classes. The unrestricted coefficient and classification of all near-minimizers remain open.

4.5 Using the actual exterior removes the shrinking amplitude rate

The loss $a^2\sqrt{A}$ in Theorem 7 charges for the entire background curvature. The chosen recovery sequence has more structure: outside the surgery cores it is close to a complex-hyperbolic metric whose full-tensor Einstein operator has a uniform positive gap. Retaining that structure removes the $h^{1/4}$ restriction.

Theorem 9 (vanishing-amplitude stability). The recovery sequence in (4.11)–(4.13), on the established balanced family with $h_j \to 0$ and $U_j = 1 + h_j + O(h_j^2)$, can be chosen so that both of the following hold:

- If k_j is exactly TT relative to g_j^0, and $\|k_j\|_\infty \to 0$, then $\liminf \mathscr{E}(\widehat{g_j^0 + k_j})/h_j \geqslant 4$.
- If k_j is exactly in the Bianchi gauge of (4.29), both endpoints have scalar curvature -4, and $\|k_j\|_\infty \to 0$, the same lower bound holds.

No relation between the amplitude and h_j, and no derivative bound or competitive-energy premise, is assumed. The gauges remain genuine hypotheses. More precisely, for $a = \|k\|_\infty \leqslant a_0$ and $x = \|\mathscr{A}_0(k)\|_2$,

where $\mathscr{A}_0$ is the norm-preserving identified derivative defined below, there are uniform positive constants c, C, a_0 such that

$$\boxed{P_{\text{final}} \geqslant P_0 + DP_0[k] + cx^2 - Ca^2Q.} \tag{4.29a}$$

The bases are selected before the perturbations. In particular the error from their near-stationarity is uniform for $a \leqslant a_0$.

Proof: the geometric decomposition. In each X_j, choose disjoint balls $B(p_i, 4R_i)$ about all surgery centers, with $N_j = k_j^{\text{surg}} + 2\ell_j \leqslant Q_j$, $R_i \leqslant r_0$, and $\sum_i R_i^4 = o(Q_j)$. Here k_j^{surg} is the blowup count, not the tensor. Reserve the seed charts of Section 4.2 separately. Choose the insertion scales much smaller than these radii. For each fixed j, the normalized metrics converge smoothly to the CH background outside the punctures. To justify this for the normalized metrics, Lemma 3 gives bounded positive Yamabe factors, their H^1 energies and convergence of the Yamabe constants. Local elliptic compactness and removability at the finitely many points produce a nonzero Yamabe minimizer on the original closed X_j. No L^4 mass is lost at the points, by the supremum and volume bounds. Uniqueness of the negative normalized representative identifies the limit with the CH metric. The diagonal can therefore ensure arbitrarily small C^2 discrepancy on the complement of $\bigcup_i B(p_i, R_i/2)$, including all forthcoming cutoff annuli. All comparisons here use the fixed CH coordinates; the exterior is not falsely declared exactly CH after normalization.

Let $C_j = \bigcup_i B(p_i, 2R_i)$ denote the corresponding regions after surgery, and $N_j^{\text{ext}} = M_j \setminus C_j$. The core curvature budget is

$$\| \operatorname{Rm}_0 \|_{L^2(C_j)} \leqslant C\sqrt{Q_j}. \tag{4.30}$$

Indeed the local Weyl integrals are conformally invariant and have bounded cost per explicit donor; the annuli and CH balls add $O(N_j) + o(Q_j)$. The Ricci energy is bounded by $P_j^0 = O(Q_j)$ and the scalar term by the core volume. The same argument applies to slightly enlarged cores. The scalar curvature of the raw construction has a uniform negative lower bound, so the Yamabe supremum estimate keeps these normalized core volumes $o(Q_j)$.

The classical exterior estimate. On every closed CH surface normalized by $\operatorname{Ric} = -g$, put $L_{\text{CH}} = \nabla^*\nabla - 2\mathring{R}$. Koiso's identity and the CH curvature action give, for every symmetric two-tensor v, not only TT tensors,

$$\langle L_{\text{CH}} v, v\rangle \geqslant \tfrac{1}{3}\|v\|_2^2, \qquad \|v\|_{H^2} \leqslant C\|L_{\text{CH}} v\|_2. \tag{4.31}$$

The constant is independent of the covering degree. See Wu, Section 3, Lemmas 3.2 and 3.6 and Proposition 3.7. In these conventions the holomorphic sectional curvature is $-2/3$; the curvature-action eigenvalues on trace, Hermitian trace-free and anti-Hermitian tensors are $-1, -1/3, 2/3$. Koiso's formula has the corresponding positive zero-order coefficients $2, 4/3, 1/3$. The H^2 bound follows by the integrated tensor Weitzenbock formula and bounded parallel curvature. Thus (4.31) does not infer a quantitative gap from qualitative local rigidity.

Localization without a gauge projection. Write $a = \|k\|_\infty$, $H = \|\nabla^2 k\|_2$, and $B_1 = \|k\|_{H^1(N^{\text{ext}})}$. For the Bianchi case define, on all symmetric tensors,

$$\mathscr{L}_0^F v = \nabla_0^*\nabla_0 v - 2\mathring{R}_0 v + Z_0 \circ v + v \circ Z_0.$$

For the TT case use $\mathscr{L}_0^Z v = \mathscr{L}_0^F v + \frac{1}{2}\langle Z_0, v\rangle g_0$. The original gauge condition gives respectively $\mathscr{L}_0^F k = 2DF_0(k)$ and $\mathscr{L}_0^Z k = 2DZ_0(k)$. Let $L_0(k)$ denote the applicable derivative $DF_0(k)$ or $DZ_0(k)$.

Splitting the bundle Weitzenbock identity (4.21) between core and exterior, using (4.30) on the core and bounded curvature outside, gives

$$H \leqslant C\{\|L_0(k)\|_2 + a\sqrt{Q} + B_1\}. \tag{4.32}$$

For detail, the curvature-square term is at most $Ca^2Q + C\|k\|^2_{2,N^{\mathrm{ext}}}$. The core curvature-gradient term is at most $C\sqrt{Q}\|\nabla k\|_4^2 \leqslant Ca\sqrt{Q}H$; the exterior term is at most CB_1^2. The rough Laplacian is then replaced by $\mathscr{L}_0 k$; its zero-order term is bounded by $Ca\sqrt{Q} + C\|k\|_{2,N^{\mathrm{ext}}}$.

Choose $\chi = 0$ on the inner balls, $\chi = 1$ outside the slightly larger balls, with transitions only in the CH-close annuli. Include every transition annulus in C_j, so that $\chi = 1$ on N_j^{ext}. Then $v = \chi k$ extends smoothly by zero to the complete covering surface X_j. The cutoff is not TT or in Bianchi gauge, and need not be: (4.31) is an all-tensor estimate, and

$$\mathscr{L}_0(\chi k) = 2\chi L_0(k) + [\nabla_0^*\nabla_0, \chi]k. \tag{4.33}$$

On its support, the small C^2 metric discrepancy gives $\|(L_{\mathrm{CH}} - \mathscr{L}_0)v\|_2 \leqslant C\epsilon_j\|v\|_{H^2}$, which is absorbed using (4.31).

The cutoff commutator has no hidden volume cost. Each annulus has $\int |\nabla^2\chi|^2 = O(1)$, so $\|k\nabla^2\chi\|_2 \leqslant Ca\sqrt{Q}$. For $w = |d\chi|^2$, integration by parts gives

$$\int w|\nabla k|^2 \leqslant Ca\sqrt{Q}\,H + Ca^2Q. \tag{4.34}$$

Indeed the Hessian term is bounded by $a\|w\|_2 H$, with $\|w\|_2 = O(\sqrt{Q})$. The other term is absorbed by Young's inequality and $|dw|^2/w \leqslant 4|\nabla^2\chi|^2$. Consequently, for any fixed $\theta > 0$,

$$\|d\chi \otimes \nabla k\|_2 \leqslant \theta H + C_\theta a\sqrt{Q}.$$

Equations (4.31)–(4.34), followed by absorption with θ small, prove the stronger graph estimate

$$\boxed{H + \|\chi k\|_{H^2(X_j, g_{\mathrm{CH}})} + \||\operatorname{Rm}_0|\,|k|\|_2 \leqslant C\{\|L_0(k)\|_2 + a\sqrt{Q}\}.} \tag{4.35}$$

The last term is the full pointwise product, not just one favorable curvature action. On the core it is at most $Ca\sqrt{Q}$; outside it is controlled by B_1.

The nonlinear conclusion. The exact Taylor remainders now satisfy

$$\|\operatorname{Rem}\|_2 \leqslant C\{aH + \|\nabla k\|_4^2 + \||\operatorname{Rm}_0|\,|k|^2\|_2\} \leqslant Ca\{\|L_0(k)\|_2 + a\sqrt{Q}\}. \tag{4.36}$$

There is no $a^2\sqrt{A}$ term. For TT input, put $b = \|s_{g^0+k} + 4\|_{2,g^0+k}$. Its linear term is $-\langle Z_0, k\rangle$, so

$$b \leqslant Ca\sqrt{Q} + Ca\{\|DZ_0(k)\|_2 + a\sqrt{Q}\}.$$

There is a stronger closure than first assuming competitive energy. Let $\mathscr{A}_0(k)$ be the identified Z-derivative in the TT case, and the identified F-derivative in the Bianchi case, using (4.16); put $x = \|\mathscr{A}_0(k)\|_2$. The identification correction is at most $Ca\sqrt{Q}$. If e is the exact remainder in the raw identified tensor, (4.36) and pointwise matrix calculus give

$$\|e\|_2 \leqslant Cax + Ca^2\sqrt{Q}, \qquad b \leqslant Ca(\sqrt{Q} + x). \tag{4.37}$$

The second inequality is needed only for the TT/Yamabe case. At the constant-scalar base, $2\langle Z_0, \mathscr{A}_0(k)\rangle = DP_0[k]$. This also holds for F, since its squared norm differs from P by $\frac{1}{4}\int(s+4)^2$. Expand the exact identified squared norm and drop $\|e\|_2^2$. In the TT case then subtract at most $b^2/12$ using (4.25); in the Bianchi case both endpoints already have $F = Z$. In either case,

$$P_{\text{final}} \geqslant P_0 + DP_0[k] + (1 - Ca)x^2 - Ca\sqrt{Q}\,x - Ca^2Q.$$

Take a_0 sufficiently small and complete the square in x. This proves (4.29a), with a positive part of x^2 retained. There was no assumption that $x = O(\sqrt{Q})$.

By (4.13), write $\|\mathrm{grad}P(g_j^0)\|_1 = \epsilon_j Q_j$, $\epsilon_j \to 0$. Then $P_{\text{final}}/Q \geqslant 8\pi^2 - Ca^2 - o(1)$, uniformly for $a \leqslant a_0$. To turn this into the relative defect, use the actual family's chiral budget

$$R_{\text{final}} - 4P_{\text{final}} \leqslant 8\pi^2 AU, \qquad \mathscr{E}_{\text{final}} = \frac{4P_{\text{final}}}{R_{\text{final}}}.$$

The function $P \mapsto 4P/(4P + 8\pi^2 AU)$ is increasing. Consequently, with $d_j = Ca_j^2 + o(1)$ chosen so $8\pi^2 - d_j > 0$,

$$\frac{\mathscr{E}_{\text{final}}}{h_j} \geqslant \frac{4(8\pi^2 - d_j)}{8\pi^2 U_j + 4h_j(8\pi^2 - d_j)} \geqslant 4 - C'a_j^2 - o(1). \tag{4.38}$$

This proves the theorem, including noncompetitive endpoints. QED. For the qualitative lower-limit assertion alone, a violating subsequence would already be competitive. The added content of (4.29a) is the uniform quadratic lower bound and its quantitative equality consequences, not merely removal of that premise.

Corollary 10 (first-order equality in these slices). Under the hypotheses of Theorem 9 with $a_j \to 0$, the following are equivalent:

$$\frac{\mathscr{E}_{\text{final}}}{h_j} \longrightarrow 4, \qquad \|\mathscr{A}_0(k_j)\|_2^2 = o(Q_j). \tag{4.39}$$

Either condition implies

$$\|\nabla_0^2 k_j\|_2 + \|\chi_j k_j\|_{H^2(X_j, g_{\mathrm{CH}})} + \||\,\mathrm{Rm}_0\,|\,k_j|\|_2 = o(\sqrt{Q_j}). \tag{4.40}$$

These are norms of the raw gauge-fixed perturbation, not a claim of uniform metric convergence after Yamabe normalization.

Proof. If $\mathscr{E}/h \to 4$, the chiral budget gives

$$P_{\text{final}} \leqslant \frac{2\pi^2 AU\mathscr{E}_{\text{final}}}{1 - \mathscr{E}_{\text{final}}} = (8\pi^2 + o(1))Q.$$

Combine with (4.29a), $P_0 = 8\pi^2 Q + o(Q)$ and $a \to 0$. Conversely, $x = o(\sqrt{Q})$ in (4.37) makes the identified remainder and scalar residual $o(\sqrt{Q})$; the exact expansion and (4.25) give $P_{\text{final}} = 8\pi^2 Q + o(Q)$. The actual scalar budget gives $V_{\text{final}} \geqslant 2\pi^2 A$, and the chiral budget gives $4V_{\text{final}} \leqslant 8\pi^2 AU + 3P_{\text{final}}$. Thus $V_{\text{final}} = 2\pi^2 A + O(Q)$, proving (4.39). Finally apply (4.35) and $\|L_0(k)\|_2 \leqslant x + Ca\sqrt{Q}$ to get (4.40). QED.

Equivalently, in these exact gauges the full raw norm $\|k_j\|_{H^2(g_j^0)}$ is $o(\sqrt{Q_j})$. Indeed the exterior L^2 norm is controlled by (4.40), while the core norm is at most $a_j\sqrt{\mathrm{Vol}(C_j)} = o(\sqrt{Q_j})$. Integration by

parts then controls the global first derivative by the zeroth and second derivatives. Conversely, this full H^2 bound, $a_j \to 0$, and the core curvature budget imply the curvature-product bound in (4.40), so the explicit gauge linearization gives $x_j = o(\sqrt{Q_j})$.

There is also low-order control of the actual final normalization, without assuming its conformal factor uniformly close to one. Write $\widehat{g} = v^2 g_{\mathrm{raw}}$, $v > 0$, and $\delta = S_{\mathrm{raw}} - S_{\widehat{g}}$. All norms in the following identities use the raw metric. The exact scalar comparison gives

$$b^2 = \delta + 16\|v^2 - 1\|_2^2 + 48\|d\log v\|_2^2, \qquad \delta = 36\|\Delta v/v\|_2^2 + 48\|dv\|_2^2. \tag{4.40a}$$

The first equality is I41's comparison written with the inverse conformal factor. For the second, substitute $s_{\mathrm{raw}} = -6\Delta v/v - 4v^2$, using the nonnegative Laplacian, square, and integrate $\int v\Delta v = \int |dv|^2$. At first-order equality, (4.37) gives $b = o(\sqrt{Q})$. Hence

$$\|v^2 - 1\|_2 + \|v - 1\|_2 + \|dv\|_2 + \|d\log v\|_2 + \|\Delta v/v\|_2 = o(\sqrt{Q}), \quad \int |v-1|^4 = o(Q). \tag{4.40b}$$

In particular $\|\widehat{g} - g_{\mathrm{raw}}\|_{2,g_{\mathrm{raw}}} = 2\|v^2 - 1\|_2 = o(\sqrt{Q})$. This is a genuine low-order normalization conclusion, not an L^∞, Hessian, or normalized-Ricci-tensor comparison. Since raw and base norms are uniformly equivalent, it also gives $\|\widehat{g} - g_j^0\|_{L^2(g_j^0)} = o(\sqrt{Q_j})$. The nonzero diffuse TT examples of Section 4.2 realize this class nontrivially. Take its bounded seeds q_j and $k_j = h_j^{3/4} q_j$. Then $x_j^2 = (c_0 + o(1))h_j^{1/2} Q_j = o(Q_j)$, with $c_0 > 0$. In (4.29a) the first-variation error is $o(h_j^{3/4} Q_j)$, and $a_j^2 Q_j = O(h_j^{3/2} Q_j)$. Thus $P_{\mathrm{final}} > P_0$ eventually: these final metrics are not merely isometric copies of the bases, yet have coefficient four. A fixed nonzero parameter instead has the positive quadratic payment in (4.10).

The equality class even allows unbounded raw curvature. Choose a compact Euclidean seed a_* whose sixth-order $DZ_{\mathrm{Eucl}}(DB_{\mathrm{Eucl}} a_*)$ is nonzero, as in Section 4.2. In each reserved chart replace it by $a_\ell(x) = \ell^4 a_*(x/\ell)$, and apply the actual Bach linearization on the locally Bach-flat normalized base. The resulting global tensors $q_{j,\ell}$ are exactly TT, $\|q_{j,\ell}\|_\infty = O(1)$, and, by four-dimensional scaling,

$$\|DZ_0(q_{j,\ell})\|_2^2 = (c_* + o(1))A_j, \qquad c_* > 0.$$

The supports are disjoint chart balls; smooth background coefficients converge after rescaling to their Euclidean principal symbols. Take $\ell_j = h_j$ and $k_j = h_j^{3/4} q_{j,\ell_j}$. Their identified derivative square is $(c_* + o(1))h_j^{1/2} Q_j = o(Q_j)$, but at a fixed rescaled seed point the raw traceless-Ricci perturbation has order $h_j^{3/4}\ell_j^{-2} = h_j^{-5/4}$. The pointwise nonlinear term is smaller by a factor $O(h_j^{3/4})$. Thus these actual raw metrics have unbounded pointwise curvature, and their normalized outputs still have first-order coefficient four and $P_{\mathrm{final}} > P_0$ by (4.29a). No pointwise curvature assertion is made for those normalized outputs. This is a test of the stated derivative-free theorem, not a new solution of the unrestricted near-minimizer problem.

More generally, competitive outputs in Theorem 9 obey the full first-order expansion

$$P_{\mathrm{final}} = 8\pi^2 Q + \|\mathscr{A}_0(k)\|_2^2 + o(Q), \qquad \frac{\mathscr{E}_{\mathrm{final}}}{h} = 4 + \frac{\|\mathscr{A}_0(k)\|_2^2}{2\pi^2 Q} + o(1), \tag{4.41}$$

because competitiveness and (4.29a) give $x = O(\sqrt{Q})$; (4.37), near-stationarity and the scalar/chiral volume bounds finish the expansion. In particular any fixed improvement $\mathscr{E}/h \leqslant 4 - \sigma$ forces $a \geqslant c\sqrt{\sigma}$ for all sufficiently large j, whenever $a \leqslant a_0$.

This last step uses the actual CH exterior and the core L^2 budget; it is stronger than merely shrinking the radius in Theorem 7. It also shows why arbitrary high derivatives do not themselves yield a sub-four competitor in these vanishing-amplitude slices. It does not provide a uniform gauge-fixing map, a fixed nonvanishing-amplitude classification, or a classification of arbitrary near-minimizers. The CH exterior is essential: on general negative KE backgrounds, TT infinitesimal Einstein deformations may exist, and (4.31) need not hold. The present theorem does not discard those genuine modes.

4.6 A measurable gauge error can replace exact Bianchi gauge

The preceding proof also yields a useful nonexact-gauge criterion. Here both actual endpoints have scalar curvature -4; this is not a claim about an arbitrary unnormalized input followed by Yamabe normalization.

Corollary 11 (amplitude-weighted gauge error). Use the selected bases of Theorem 9. Let $g_j = g_j^0 + k_j$ have scalar curvature -4, and put

$$a_j = \|k_j\|_{\infty,g_j^0}, \qquad G_j = \|\delta_0^*\beta_0 k_j\|_{2,g_j^0}, \qquad \beta_0 k = \delta_0 k + \tfrac{1}{2} d\,\mathrm{tr}_0\, k.$$

If $a_j \to 0$ and $a_j G_j = o(\sqrt{Q_j})$, then $\liminf_j \mathscr{E}(g_j)/h_j \geqslant 4$. There is no competitive-energy premise. For $a \leqslant a_0$, the quantitative bound, with $x = \|\mathscr{A}_0^F(k)\|_2$, is

$$\boxed{P_g \geqslant P_0 + DP_0[k] + cx^2 - C\{a^2Q + aG\sqrt{Q} + a^2G^2\}.} \tag{4.42}$$

Proof. With $\delta = -\,\mathrm{div}$ and δ^* the symmetrized covariant derivative, the exact linearization is

$$DF_0(k) = \tfrac{1}{2}\mathscr{L}_0^F k - \delta_0^*\beta_0 k.$$

Repeating the cutoff proof with this explicit source gives

$$\|\nabla^2 k\|_2 + \|\chi k\|_{H^2(X_j,g_{\mathrm{CH}})} + \|\,|\mathrm{Rm}_0|\,k\|_2 \leqslant C\{\|DF_0(k)\|_2 + G + a\sqrt{Q}\}. \tag{4.43}$$

Since the identified derivative differs by at most $Ca\sqrt{Q}$, the exact identified remainder satisfies

$$\|e\|_2 \leqslant Cax + CaG + Ca^2\sqrt{Q}.$$

At both endpoints $F = Z$. Expand its squared norm as before. The terms involving x are absorbed into a fixed fraction of x^2; the others are bounded by the last term of (4.42). This proves the finite inequality. The stated asymptotic conditions make that last term $o(Q)$; near-stationarity and the monotone chiral conversion (4.38) prove the conclusion. QED.

In this larger class, the equivalence $\mathscr{E}/h \to 4$ if and only if $x^2 = o(Q)$ also holds. However the unweighted derivative conclusion (4.40) need not follow: G itself need not be $o(\sqrt{Q})$. Its amplitude-weighted size, not its full norm, controls the nonlinear energy loss.

Constant scalar curvature does not supply this gauge condition. For example, on the reserved uniformly controlled charts take diffeomorphisms equal to the identity near their boundaries, with the local shear

$$\phi_j(x) = (x_1, x_2 + (a_j/\lambda_j)\chi(x)\sin(\lambda_j x_1), x_3, x_4).$$

Replicate over the lifted charts and put $g_j = \phi_j^* g_j^0$. These are actual scalar-minus-four metrics with exactly the same $P, V, \mathscr{E}$ as the base. Their difference has supremum norm comparable to a_j, whereas on $\chi = 1$ the principal term of $\delta^*\beta k$ has size $a_j\lambda_j^2/2$. Consequently G_j is bounded below by a constant times $a_j\lambda_j^2\sqrt{A_j}$, after taking the frequencies large enough to dominate the bounded-coordinate lower-order errors. Thus even $a_j G_j/\sqrt{Q_j}$ can diverge. This is pure coordinate motion, not a lower-energy geometry: the inverse diffeomorphism restores the exact base. It shows why a uniform gauge-fixing result, rather than a scalar or energy estimate alone, is required to pass to arbitrary nearby metrics modulo diffeomorphisms.

The weighted condition is genuinely weaker than $G = o(\sqrt{Q})$. In the same pullback construction choose $a_j = h_j$ and $\lambda_j = h_j^{-2/3}$. Then

$$G_j/\sqrt{Q_j} \asymp h_j^{-5/6} \longrightarrow \infty, \qquad a_j G_j/\sqrt{Q_j} \asymp h_j^{1/6} \longrightarrow 0.$$

These actual representatives satisfy Corollary 11 and retain the base energy exactly. This check uses diffeomorphism invariance, not a new family of lower-energy metrics.

Nonconstant-scalar inputs. One may also relax TT before the final Yamabe normalization, provided the additional scalar-linear error is retained. For a general smooth k, define

$$D = \|Ds_0[k]\|_2, \qquad G = \|\delta_0^*\beta_0 k\|_2, \qquad x = \|\mathscr{A}_0^Z(k)\|_2.$$

For $a = \|k\|_\infty \leqslant a_0$, the same proof gives

$$\boxed{P_{\widehat{g_0+k}} \geqslant P_0 + DP_0[k] + cx^2 - C\{a^2Q + aG\sqrt{Q} + a^2G^2 + D^2\}.} \tag{4.44}$$

Indeed $DF_0(k) = DZ_0(k) + \frac{1}{4}Ds_0[k]g_0$. The graph estimate therefore has right side $C(x + D + G + a\sqrt{Q})$. Its raw identified remainder is bounded by $Ca(x + D + G) + Ca^2\sqrt{Q}$, and its scalar residual by $D + Ca(x + D + G + a\sqrt{Q})$. Subtract (4.25), complete the squares in x, and use $aD\sqrt{Q} \leqslant (a^2Q + D^2)/2$. This proves (4.44). Consequently the conditions

$$a_j \to 0, \qquad a_j G_j = o(\sqrt{Q_j}), \qquad D_j = o(\sqrt{Q_j})$$

are sufficient for the coefficient-four lower bound, with equality if and only if $x_j^2 = o(Q_j)$. At first-order equality the low-order final-factor conclusions (4.40a)–(4.40b) persist; they are not asserted for arbitrary noncompetitive outputs. Exact TT is included because $Ds_0[k] = -\langle Z_0, k\rangle$. Neither the scalar-linear nor the weighted gauge condition has been proved automatic for arbitrary small-amplitude inputs.

Only the trace-free gauge forcing is needed in these two estimates. Precisely, G can be replaced in (4.42) and (4.44) by $G^\circ = \|(\delta^*\beta_0 k)^\circ\|_2$. This is a refinement of the same criterion, not another gauge-existence claim. Here is the trace calculation, to specify why no scalar direction has been suppressed. Write $t = \mathrm{tr}_0 k$ and $q = k - tg_0/4$. The exact identities are

$$DZ_0[k] = \tfrac{1}{2}\mathscr{L}_0^F q - (\delta^*\beta_0 k)^\circ + \tfrac{1}{4}\langle Z_0, k\rangle g_0, \qquad Ds_0[k] = (\tfrac{3}{4}\Delta + 1)t + \delta\delta q - \langle Z_0, q\rangle. \tag{4.45}$$

Apply the patched full-tensor estimate to q with the displayed source. This bounds its graph norm by $C(x_Z + G^\circ + a\sqrt{Q})$. The positive scalar operator in (4.45) then gives

$$\|t\|_2 + \|\Delta t\|_2 \leqslant C(D + \|\nabla^2 q\|_2 + a\sqrt{Q}).$$

For its Hessian use scalar Bochner with $\mathrm{Ric}_0 = Z_0 - g_0$, $\|Z_0\|_2 = O(\sqrt{Q})$, and $\|dt\|_4^2 \leqslant Ca\|\nabla^2 t\|_2$. Young's inequality costs only Ca^2Q, not Ca^2A. The trace curvature-product norm is bounded on the core by $Ca\sqrt{Q}$ and on the bounded-curvature exterior by $C\|t\|_2$. Thus the complete graph norm is at most $C(x_Z + G^\circ + D + a\sqrt{Q})$. For constant-scalar endpoints the exact scalar expansion gives $D \leqslant Ca$ times this graph norm. If the F-identification is used, $x_Z \leqslant x_F + D/2$, so the D term still absorbs for small a. The previous remainder and square-completion proofs now apply with G°, retaining D^2 only in the general raw/Yamabe case.

5. Exact scope of this continuation

The weak zero endpoint still requires an additional differential or global implication. Equations (1.4) and (2.5) identify a weighted derivative control and the specific covariance which obstructs an unweighted conclusion; they do not dispose of it by interpolation. The existing small local cores are not closed actual escape sequences.

The construction-side conclusion is complete in its defined class: arbitrary backgrounds and subsequent Yamabe minimization have the limits (3.2), (3.5), with the sharp value (3.6), penalties (3.10)–(3.11), and perturbative escape threshold (4.4)–(4.5). The conformal tensor lemma and (4.10) supply a nonzero actual test of the diffuse variation. Proposition 6 then treats core directions on the specified almost-critical recovery sequence, and (4.17) measures the nonlinear change required of any successful path there. Theorem 7 and Corollary 8 remove the assumed Taylor control in two explicit small-amplitude gauges by absorbing the actual competitive energy. Theorem 9 uses the classical all-tensor CH gap and actual cutoff/core geometry to remove the h-dependent amplitude rate and the competitive-energy premise. Its direct nonlinear inequality retains a positive quadratic term; Corollary 10 identifies the first-order equality condition within the specified actual slices. Corollary 11 permits amplitude-weighted gauge error on actual constant-scalar endpoints; it does not automatically place other metrics in that class. I44 already supplies the underlying unchanged-cap obstruction. These deductions constrain first-order geometry but do not determine the all-metric theorem's coefficient.

The next unresolved implications are therefore concrete: control the actual correlation and the subsequent torsion-length derivative relation, or build a closed zero-end sequence; and on the balanced family, control the full signed Ah-scale remainder or realize geometry beyond the construction/perturbation class above. No additional field supplier, global core decomposition, or all-metric optimality is asserted. For the new stability route the missing intrinsic step is a uniform amplitude-preserving gauge reduction through the shrinking donors. Its vector-field linearization is $\beta_0(\mathscr{L}_X g_0) = (\nabla^*\nabla - \mathrm{Ric}_0)X^\flat$. The CH symmetric-tensor gap used here does not itself solve that global nonlinear gauge problem.